Exp 6 · CE amplifier: frequency response and bandwidth

Manual pp. 20–21 · hands-on · CO4, CO5 · the same stage is reused in the extra experiment

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What you already have: from Exp 5, in the active region $I_C = \beta I_B$, and the BE junction needs ≈ 0.7 V. From C&S, a capacitor's reactance $X_C = 1/(2\pi fC)$: huge at low f, tiny at high f.

Three terms first:

- **Biasing:** DC voltages that park the transistor in the active region at a chosen **Q-point** (quiescent point = the DC operating point with no signal).
- **Small-signal gain A_v:** v_o/v_{in} for a small sine riding on the Q-point. In decibels: $20 \log_{10} A_v$.
- **Bandwidth:** the frequency range over which the gain stays within 3 dB (a factor of 0.707) of its maximum.

MANUAL EXP 6, AIM

To plot the frequency response of a CE amplifier and determine its bandwidth. Viva form: "Why does the gain fall at **both** low and high frequencies, and what sets each cut-off?"

GUESS FIRST

Name one part responsible for each end before reading on.

The circuit, part by part

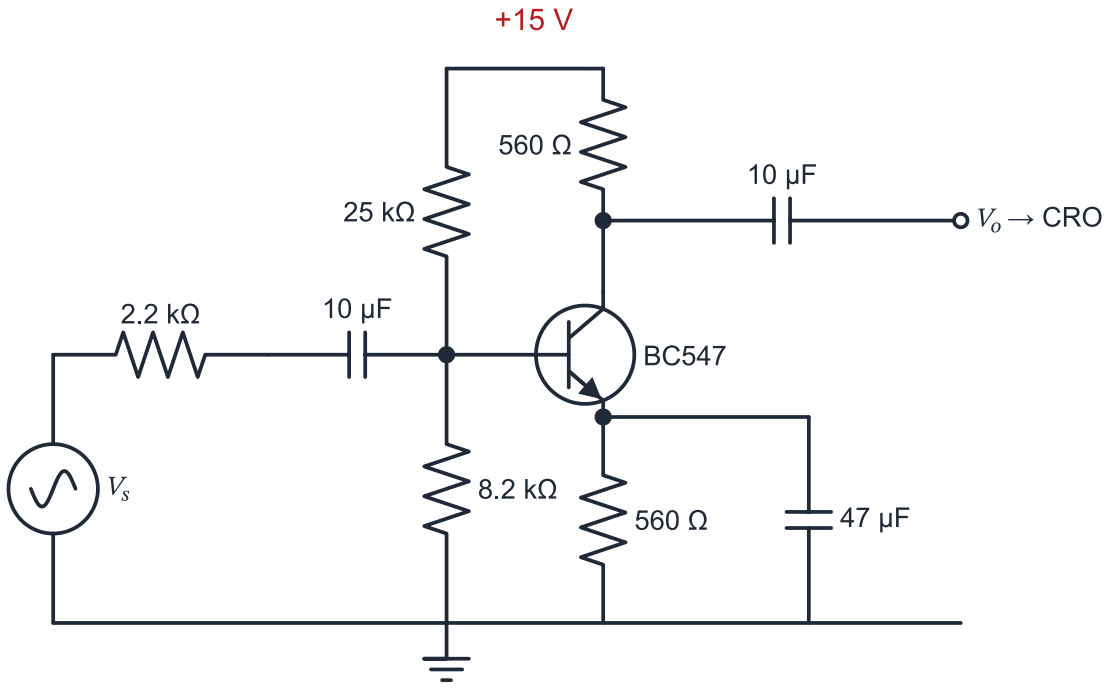


Fig 6.1, redrawn: voltage-divider bias, emitter resistor with bypass capacitor, coupling capacitors in and out.

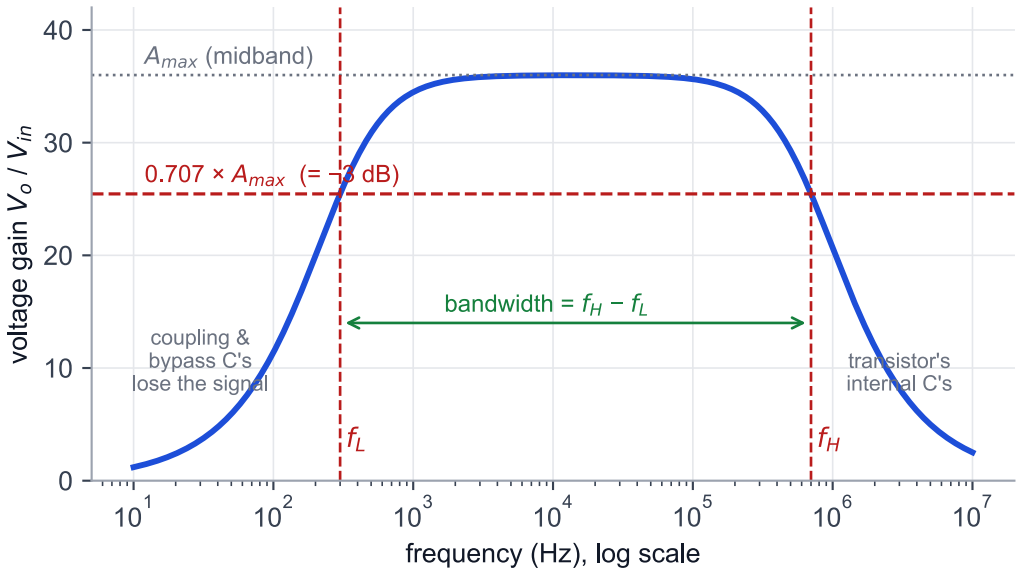
Part	Job
25 kΩ + 8.2 kΩ (R1, R2)	voltage divider: fixes $V_B \approx 15 \times 8.2/33.2 \approx 3.6$–$3.7$ V
560 Ω emitter (R_E)	sets $I_E = V_E/R_E$. It also stabilises the bias: if I_C rises with temperature, V_E rises, which reduces V_{BE}.
47 µF across R_E (C_E, **bypass**)	shorts R_E for AC, so the signal sees the emitter grounded. This gives full gain.
10 µF input (C_{in}, **coupling**)	passes the signal, blocks DC (so the generator doesn't upset the bias)
10 µF output (C_C)	passes the amplified signal, blocks the 12 V collector DC from the CRO
560 Ω collector (R_C)	turns the collector current change into a voltage change

Q-point (LTspice .op): $V_B = 3.60$ V, $V_E = 2.92$ V, $I_C = 5.2$ mA, $V_C = 12.09$ V, so $V_{CE} = 9.2$ V. The transistor is safely active.

Gain by hand:

- r_e = 26 mV / I_E = 26/5.2 = **5 Ω**. This is the BE junction's small-signal resistance.
- Base to collector: $A_v = -R_C/r_e = -560/5 = -112$.
- The 2.2 kΩ source resistor and the amplifier's input resistance (R_{in} = 25k ∥ 8.2k ∥ βr_e ≈ 1.2 kΩ) form a divider. Only 1.2/(1.2 + 2.2) = 0.35 of v_s reaches the base.
- Overall: $v_o/v_s \approx 0.35 \times 112 \approx$ **39**. LTspice gives **36**.
- The minus sign means the output is **inverted** (180° phase shift).

Why the gain falls at both ends



The general shape. Low end: capacitor reactances grow. High end: the transistor's internal junction capacitances short the signal.

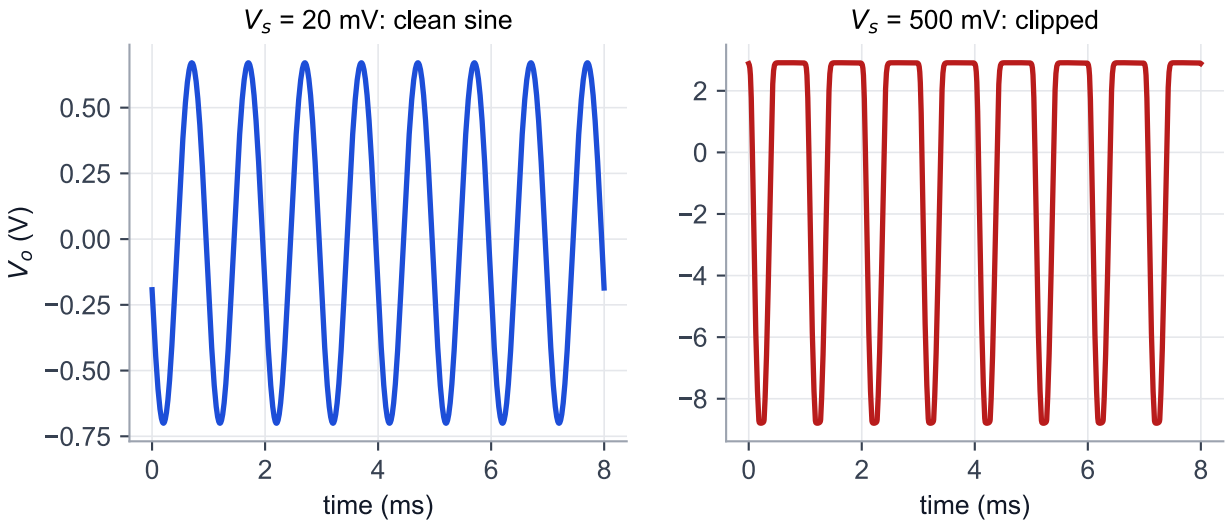
- **Low f:** $X_C = 1/(2\pi fC)$ grows. The coupling capacitors start to block the signal, and above all **C_E stops bypassing R_E**, which brings in negative feedback (see Exp 8). The lower cut-off is set mostly by C_E: $f_L \approx 1/(2\pi\, C_E\, R_e')$, where $R_e' \approx r_e$ + (source ∥ bias)/β ≈ 10.6 Ω. That gives 1/(2π × 47 µF × 10.6 Ω) ≈ **320 Hz**.
- **High f:** tiny capacitances inside the transistor (between its junctions) and in the wiring start to short the signal. The base–collector capacitance is multiplied by the gain (the *Miller effect*), so it dominates.
- **Middle (midband):** all external capacitors act as shorts and the internal ones as opens, so the gain is flat.

DO IT (HANDS-ON)

1. Build Fig 6.1 on the breadboard with supply **+15 V**. 25 kΩ is not a standard value: use 22 kΩ + 3.3 kΩ in series. Check the electrolytic capacitors' polarity: + towards the higher DC voltage (base side for C_{in}, collector side for C_C, emitter side for C_E).
2. **Check the Q-point first** with the DMM, no signal applied: $V_B \approx 3.6$ V, $V_E \approx 2.9$ V, $V_C \approx 12$ V. If $V_C \approx 15$ V the transistor is off; if $V_C \approx V_E$ it is saturated. Fix this before going on.
3. Generator: sine, 1 kHz. Set the amplitude **small: about 20 mV peak** (use the generator's −20 dB/attenuator button, or a 1 kΩ:100 Ω divider). Look at V_o on the DSO. It must be a clean sine, not flattened.
4. Keep V_s constant. Step the frequency through **100 Hz, 500, 800, 1k, 5k, 10k, 50k, 100k, 500k, 800k, 1 MHz** and read $V_{o(pp)}$ at each.
5. Calculate $A_v = V_o/V_s$ and 20 log A_v. Plot dB (y) against frequency on **semi-log** paper (log x-axis).
6. Draw a line 3 dB below the maximum. Where it cuts the curve gives f_L and f_H. BW = $f_H - f_L$.

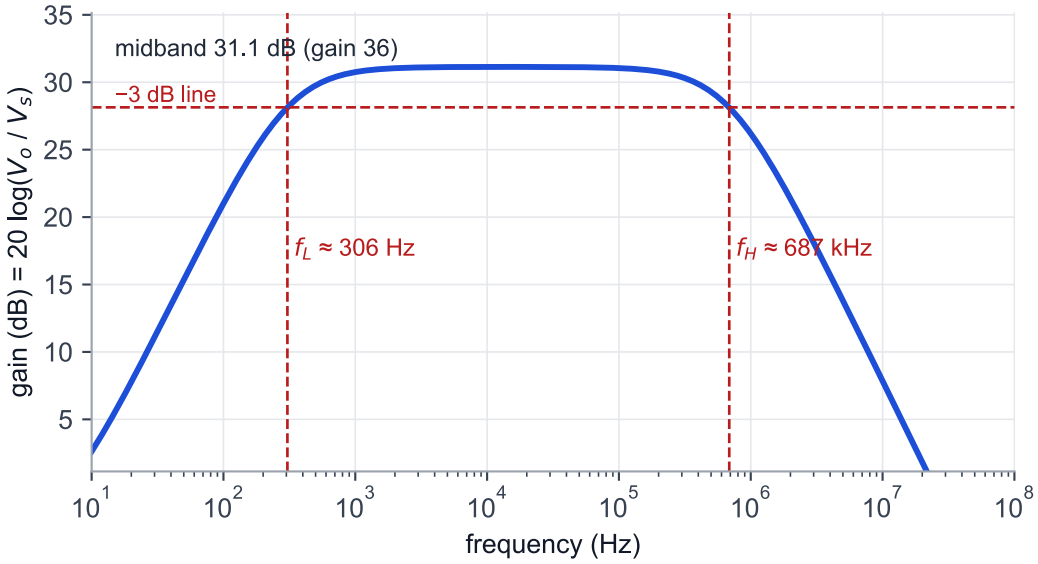
TRAP: THE MANUAL'S 500 MV INPUT

At 500 mV the output would need to swing about ±18 V, but the collector can only move between ≈ 3 V and 15 V. The output is clipped flat (below), so the "gain" you read is meaningless. Even the 100 mV in Fig 6.1 distorts (+2.5 / −3.4 V). Use about 20 mV.



LTspice: the same amplifier with 20 mV in (clean, gain ≈ 34) and 500 mV in (clipped).

What you should see



LTspice, Fig 6.1 with the CRO as a 1 MΩ load. Midband gain 36 (31.1 dB), $f_L \approx 306$ Hz, $f_H \approx 687$ kHz.

f (Hz)	100	500	800	1k	5k	10k	50k	100k	500k	800k	1M
$A_v = V_o/V_s$	11.2	30.8	33.7	34.5	36.0	36.1	36.0	35.7	29.2	23.5	20.4
gain (dB)	21.0	29.8	30.6	30.8	31.1	31.1	31.1	31.1	29.3	27.4	26.2

On the bench, f_L should land close to 300 Hz, because it is set by the capacitors you placed. f_H depends on breadboard wiring, probe and generator capacitance, so expect anything from a few hundred kHz upward. The simulation is the ideal-wiring value.

Calculation

$$BW = f_H - f_L = 687\ \text{kHz} - 0.306\ \text{kHz} \approx 687\ \text{kHz}$$

When $f_H \gg f_L$, the bandwidth is essentially f_H.

WRITE IN YOUR RECORD

Result: maximum gain ≈ **36 (31.1 dB)**; $f_L \approx$ **306 Hz**, $f_H \approx$ **687 kHz**; bandwidth ≈ **687 kHz** (simulation; use your measured values).

Theory for the record: in a CE amplifier the input is applied at the base and the output taken at the collector, with the emitter common (grounded for AC by the bypass capacitor). Voltage-divider bias with an emitter resistor sets a stable Q-point in the active region. A small base signal produces a β-times-larger collector current change, which develops a large, phase-inverted voltage across R_C. Gain ≈ R_C/r_e. The frequency response is flat in midband, falls at low frequencies because of the coupling and bypass capacitors, and falls at high frequencies because of junction and stray capacitances. Bandwidth = $f_H - f_L$, measured at the −3 dB (0.707 A_{max}) points, also called the half-power points.

CHECK 1

(a) Why is the −3 dB point called the "half-power" point? (b) You remove C_E. Does the gain go up or down, and roughly to what? (c) V_C reads 15 V with no signal. What is wrong?

Answers

Opening question. Low end: the **coupling and bypass capacitors**' reactance becomes large; C_E matters most ($f_L \approx 1/(2\pi C_E R_e')$). High end: the **transistor's internal junction capacitances** (Miller-multiplied C_{bc}) plus stray capacitance.

Check 1. (a) Power $\propto V^2$. At 0.707 of the voltage, the power is $0.707^2 = 0.5$ of maximum, and $20 \log 0.707 = -3$ dB. (b) **Down.** R_E returns to the AC path, so $A_v \approx -R_C/(r_e + R_E) = -560/565 \approx -1$. The overall gain is about 1 or less (this is Exp 8's effect). (c) No collector current: the transistor is cut off. Look for an open base connection, a missing emitter return, a wrong pin order (check the BC547's pinout: C-B-E with the flat face towards you) or a dead transistor.